

Can't wait to myelinate.

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Oligodendrocytes myelinate axons in the central nervous system (CNS). In this issue of *Developmental Cell*, Czopka et al. (2013) shed light on the temporal control of myelination by individual cells. They demonstrate that oligodendrocytes in vivo have only a brief time window to initiate myelination, which has important implications for CNS plasticity.

Biology of oligodendrocyte and myelin in the mammalian central nervous system.

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Oligodendrocytes, the myelin-forming cells of the central nervous system (CNS), and astrocytes constitute macroglia. This review deals with the recent progress related to the origin and differentiation of the oligodendrocytes, their relationships to other neural cells, and functional neuroglial interactions under physiological conditions and in demyelinating diseases. One of the problems in studies of the CNS is to find components, i.e., markers, for the identification of the different cells, in intact tissues or cultures. In recent years, specific biochemical, immunological, and molecular markers have been identified. Many components specific to differentiating oligodendrocytes and to myelin are now available to aid their study. Transgenic mice and spontaneous mutants have led to a better understanding of the targets of specific dys- or demyelinating diseases. The best examples are the studies concerning the effects of the mutations affecting the most abundant protein in the central nervous myelin, the proteolipid protein, which lead to dysmyelinating diseases in animals and human (jimpy mutation and Pelizaeus-Merzbacher disease or spastic paraplegia, respectively). Oligodendrocytes, as astrocytes, are able to respond to changes in the cellular and extracellular environment, possibly in relation to a glial network. There is also a remarkable plasticity of the oligodendrocyte lineage, even in the adult with a certain potentiality for myelin repair after experimental demyelination or human diseases.

Biogenesis and significance of central nervous system myelin.

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The central nervous system is composed of neurons and glia cells. Although neurons have long been considered the functionally important cells, an ever-expanding body of research has revealed many critical functions of neuroglia. Among these, the myelin sheath elaborated by oligodendrocytes acts as a dynamic partner to the axons it enwraps and can no longer be considered as an inert membrane. In addition to its best known roles of providing insulation and optimizing conduction velocity, myelination modulates the maturation, survival, and regenerative capacity of axons through trophic support and signaling molecules. Myelin is produced through a complex process involving cell differentiation, biosynthesis of specialized lipids and proteins, interaction with environmental signals, and coordinated changes in cell morphology. Understanding the pathophysiology of primary myelin disorders, and the challenges faced in treating them, is facilitated through understanding of the structure, function, and generation/regeneration of myelin.

Neuronal activity and NIBS in developmental myelination and remyelination - Current state of knowledge.

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Oligodendrocytes are responsible for myelinating central nervous system (CNS) axons and rapid electrical transmission through saltatory conduction of action potentials. Myelination and myelin repair rely partially on oligodendrogenesis, which comprises oligodendrocyte precursor cell (OPC) migration, maturation, and differentiation into oligodendrocytes (OL). In multiple sclerosis (MS), demyelination occurs due to an inflammatory cascade with auto-reactive T-cells. When oligodendrogenesis fails, remyelination becomes aberrant and conduction impairments are no longer restored. Although current disease modifying therapies have achieved results in modulating the faulty immune response, disease progression continues because of chronic inflammation, neurodegeneration, and failure of remyelination. Therapies have been tried to promote remyelination. Modulation of neuronal activity seems to be a very promising strategy in preclinical studies. Additionally, studies in people with MS (pwMS) have shown symptom improvement following non-invasive brain stimulation. (NIBS) techniques. The aforementioned mechanisms are yet unknown and probably involve both the activation of neurons and glial cells. Noting neuronal activity contributes to myelin plasticity and that NIBS modulates neuronal activity; we argue that NIBS is a promising research horizon for demyelinating diseases. We review the hypothesized pathways through which NIBS may affect both neuronal activity in the CNS and how the resulting activity can affect oligodendrogenesis and myelination.

Axotomy-induced cytoskeleton changes in unmyelinated mammalian central nervous system axons.

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Oligodendrocyte-derived myelin retards the ability of CNS axons to regenerate following transection. The intrinsic response of CNS axons to an axotomy insult may be vastly different in the absence of myelin. However, the paucity of adequate experimental models has limited detailed investigation of cellular behaviour following axon transection in an unmyelinated CNS environment. In this study we perform laser-induced axotomy of the porcine retinal ganglion cell axon, a physiologically unmyelinated, mature CNS axon that is structurally similar to humans to infer knowledge about axonal behaviour in the absence of myelin. Axotomy-induced changes to the neuronal cytoskeleton and supporting astrocytes during the early stages after transection are delineated by examining the sequence of neurofilament subunit, microtubule (TUB), microtubule associated protein (MAP), glial fibrillary acidic protein (GFAP) and terminal deoxynucleotidyl transferase biotin-dUTP nick end labelling (TUNEL) modification. Axonal transection induced an increase in the expression of neurofilament light at regions within, and immediately adjacent to, sites of axotomy. Other neurofilament subunits were not altered at sites of transection. Unlike myelinated axons where an increase in GFAP staining within hypertrophic glial scars have been shown to inhibit axonal repair we demonstrate a decrease in GFAP staining within regions of increased or preserved neurofilament expression. The behaviour of TUB and MAP proteins following transection of unmyelinated CNS axons are similar to what has previously been described in myelinated CNS axons. This study provides fundamental insights into astrocyte and axonal behaviour acutely after axotomy and demonstrates a series of degenerative events in unmyelinated CNS axons, which in comparison to prior reports are different to myelinated CNS axons. The findings of this report have relevance to understanding pathogenic mechanisms underlying neuro-degeneration in the CNS.

Effects of ascorbic acid on myelination in offspring of advanced maternal age.

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Myelination is the process by which oligodendrocytes ensheath axons to form myelin sheaths. Myelination is a crucial aspect of brain development and is closely associated with central nervous system abnormalities. However, previous studies have found that advanced maternal age might affect the myelination of offspring, potentially through the pathway of disrupting DNA methylation levels in the offspring's hippocampus. Current research has demonstrated that ascorbic acid can promote hydroxymethylation to reduce methylation levels in vivo. This study aims to verify the relationship between ascorbic acid and myelination, as well as the specific mechanism involved. Initially, oligodendrocyte differentiation was observed using immunofluorescence and Western blot. Myelination was assessed through Luxol Fast Blue staining, Glycine silver staining, immunofluorescence, and transmission electron microscopy. The demethylation level of oligodendrocyte progenitor cells was detected by immunofluorescence co-expression of OLIG2 and DNA hydroxylase ten-eleven translocation 1 (TET1), TET2, and TET3. Our study found that advanced maternal age could impair myelination in the hippocampus and corpus callosum of offspring. Ascorbic acid intervention may induce TET1 and TET2-mediated hydroxymethylation to ameliorate myelination disorders, promote myelination and maturation, and reverse the effects of advanced maternal age on offspring.

A genetic screen identifies genes essential for development of myelinated axons in zebrafish.

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The myelin sheath insulates axons in the vertebrate nervous system, allowing rapid propagation of action potentials via saltatory conduction. Specialized glial cells, termed Schwann cells in the PNS and oligodendrocytes in the CNS, wrap axons to form myelin, a compacted, multilayered sheath comprising specific proteins and lipids. Disruption of myelinated axons causes human diseases, including multiple sclerosis and Charcot-Marie-Tooth peripheral neuropathies. Despite the progress in identifying human disease genes and other mutations disrupting glial development and myelination, many important unanswered questions remain about the mechanisms that coordinate the development of myelinated axons. To address these questions, we began a genetic dissection of myelination in zebrafish. Here we report a genetic screen that identified 13 mutations, which define 10 genes, disrupting the development of myelinated axons. We present the initial characterization of seven of these mutations, defining six different genes, along with additional characterization of mutations that we have described previously. The different mutations affect the PNS, the CNS, or both, and phenotypic analyses indicate that the genes affect a wide range of steps in glial development, from fate specification through terminal differentiation. The analysis of these mutations will advance our understanding of myelination, and the mutants will serve as models of human diseases of myelin.

Endoplasmic reticulum stress and the unfolded protein response in disorders of myelinating glia.

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Myelin is vital to the proper function of the nervous system. Oligodendrocytes in the CNS and Schwann cells in the PNS are the glial cells responsible for generating the myelin sheath. Myelination requires the production of a vast amount of proteins and lipid-rich membrane, which puts a heavy load on the secretory pathway of myelinating glia and leaves them susceptible to endoplasmic reticulum (ER) stress. Cells respond to ER stress by activating the unfolded protein response (UPR). The UPR is initially protective but in situations of prolonged unresolved stress the UPR can lead to the apoptotic death of the stressed cell. There is strong evidence that ER stress and the UPR play a role in a number of disorders of myelin and myelinating glia, including multiple sclerosis, Pelizaeus-Merzbacher disease, Vanishing White Matter Disease, and Charcot-Marie-Tooth disease. In this review we discuss the role that ER stress and the UPR play in these disorders of myelin. In addition, we discuss the progress that has been made in our understanding of the effect genetic and pharmacological manipulation of the UPR has in mouse models of these disorders and the novel therapeutic potential of targeting the UPR that these studies support. This article is part of a Special Issue entitled SI:ER stress.

The pathogenesis of demyelinating disease: insights from cell biology.

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Cellular and humoral immune mechanisms have been implicated in the pathogenesis of human and experimental demyelinating diseases of the CNS. How these interact in the complex sequence of events that culminates in phagocytosis of myelin by macrophages has yet to be resolved. The relationship between leakage of the blood-brain barrier and demyelination, the reason why recurrent inflammatory demyelination occurs--seemingly in the absence of an antigen-specific immune response--and the lack of effective remyelination all require explanation if a coherent account of immunologically mediated demyelination is to be achieved. One approach to these problems is to study in vitro the developmental and cellular biology of oligodendrocytes--the glial cells responsible for the synthesis and maintenance of CNS myelin. This provides experimental opportunities not offered by more direct investigation of the intact nervous system, but carries the clear disadvantage that observations made in vitro cannot necessarily be extrapolated to humans.

Cytokines and myelination in the central nervous system.

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Myelin abnormalities that reflect damage to developing and mature brains are often found in neurological diseases with evidence of inflammatory infiltration and microglial activation. Many cytokines are virtually undetectable in the uninfamed central nervous system (CNS), so that their rapid induction and sustained elevation in immune and glial cells contributes to dysregulation of the inflammatory response and neural cell homeostasis. This results in aberrant neural cell development, cytotoxicity, and loss of the primary myelin-producing cells of the CNS, the oligodendrocytes. This article provides an overview of cytokine and chemokine activity in the CNS with relevance to clinical conditions of neonatal and adult demyelinating disease, brain trauma, and mental disorders with observed white matter defects. Experimental models that mimic human disease have been developed in order to study pathogenic and therapeutic mechanisms, but have shown mixed success in clinical application. However, genetically

altered animals, and models of CNS inflammation and demyelination, have offered great insight into the complexities of neuroimmune interactions that impact oligodendrocyte function. The intracellular signaling pathways of selected cytokines have also been highlighted to illustrate current knowledge of receptor-mediated events. By learning to interpret the actions of cytokines and by improving methods to target appropriate predictors of disease risk selectively, a more comprehensive understanding of altered immunoregulation will aid in the development of advanced treatment options for patients with inflammatory white matter disorders.
